



Online Auction & Bidding Platform

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Abstract

- Web platform for listings, auctions, live bidding and secure transactions. Stack: React, Node/Express, MySQL, Redis, Socket.IO, Stripe and JWT.
- Outcome: secure accounts, real-time bids and automated processing.

Problem Statement & Objectives

- Problem: Basic systems lack live bidding, secure access, independent services and automation.
- Objectives: secure management.
- Socket.IO updates; JWT authentication.
- Independent microservices.
- Scope: users, products, auctions, bids, payments and dashboards.

Existing vs Proposed System

Feature	Existing System	Proposed System
Auth / Security	Basic auth	JWT auth and protected APIs
Backend / Data	Monolith; centralized	Microservices; MySQL + Redis services
Bidding / Auctions	Delayed; basic handling	Live Socket.IO updates; timers and automated lifecycle
Payment	Manual/external	Stripe integration
Security	Limited access	Token-based control

Feasibility Study

- Technical: React, Node/Express, JWT, MySQL, Redis, Socket.IO and Stripe are available.
- Economic: Open-source tools reduce implementation cost.
- Operational: Simple registration, auctions, bidding and transaction management.
- Schedule: Design, build and testing fit the semester PBL timeline.

Results & Testing

- Tested: registration/login, JWT, protected APIs, CRUD, bid validation, Socket.IO, timers, Redis, payments and microservices.

Conclusion

- Delivered secure online auctions with live bidding.
- Uses JWT, microservices, REST, MySQL, Redis, Socket.IO and Stripe.

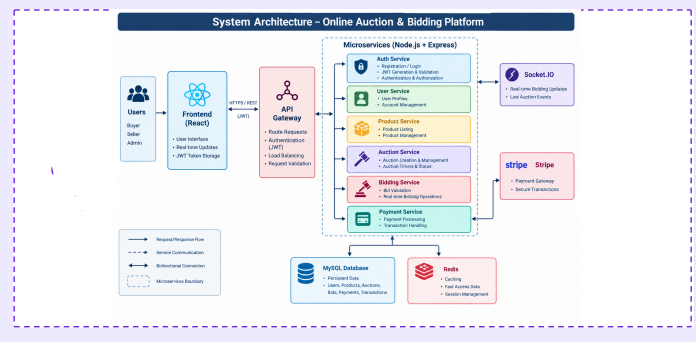
Future Scope

- AI recommendations, personalization and fraud detection.
- Mobile support, cloud scaling and notifications.

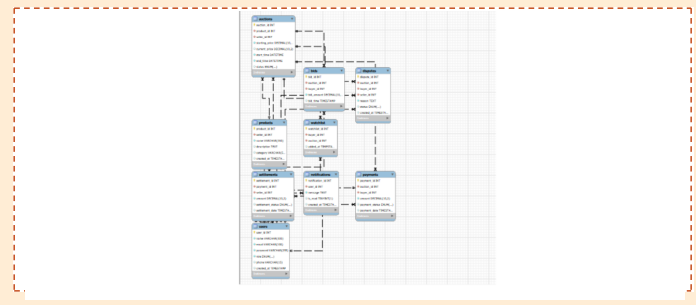
Tools & Technologies

- Front end: React, HTML, CSS, JavaScript
- Back end: Node/Express, REST APIs, microservices
- Database: MySQL | Cache: Redis | Auth: JWT
- Live: Socket.IO | Payment: Stripe | Tools: GitHub, Postman, VS Code

System Architecture



ER Diagram & Database Design



Implementation Screenshots

